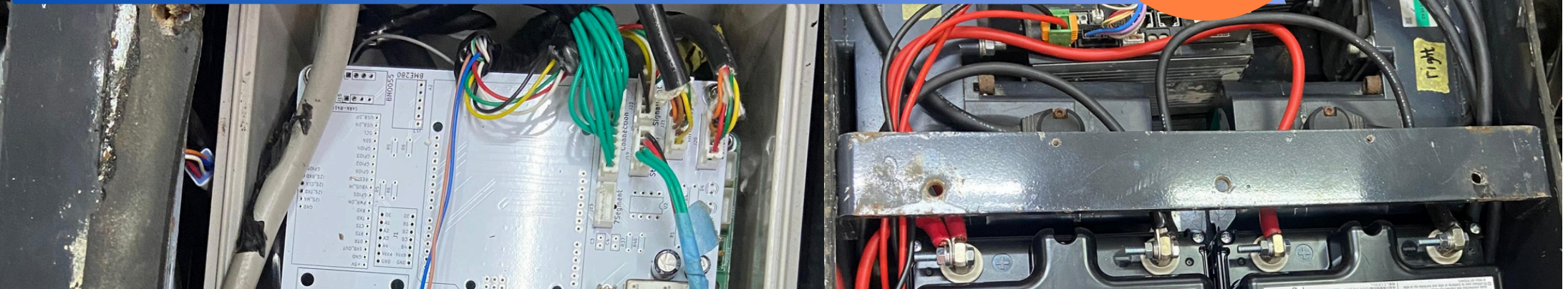


WIRING DIAGRAM DOCUMENTATION – MSD700

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Base System & Attachment Arm
Wiring Reference

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DISCLAIMER & DOCUMENT SCOPE

Disclaimer:

This wiring documentation is prepared based on cable tracing, direct observation, and direct inspection of the MSD700

Therefore, the information presented here is intended for reference, understanding, and maintenance purposes only, and better not be used as an official design or manufacturing specification

This document covers:

- General system-level wiring overview
- Detailed wiring of the Base MSD700 system
- Detailed wiring of the Attachment Arm subsystem
- wiring diagram of the charger module interface (Indonesia power Grid)
- Reference to relevant datasheets and operational notes





INTRODUCTION

1 Purpose of This Document

This document provides a structured wiring reference for the MSD700 system. It is intended to help understand the interconnection between major electrical components, including power distribution, motor drivers, actuators, safety devices, and indicator systems.

The documentation is designed to be very easy to read while still maintaining sufficient technical detail for maintenance, troubleshooting, and system review

2 How to Read the document

This document is organized in a hierarchical manner, starting from a high-level overview and gradually moving toward more detailed wiring information.

- The initial sections present block diagrams that show the overall system architecture.
- Subsequent sections provide detailed wiring diagrams for each subsystem.

To improve readability and navigation (please download the file first):

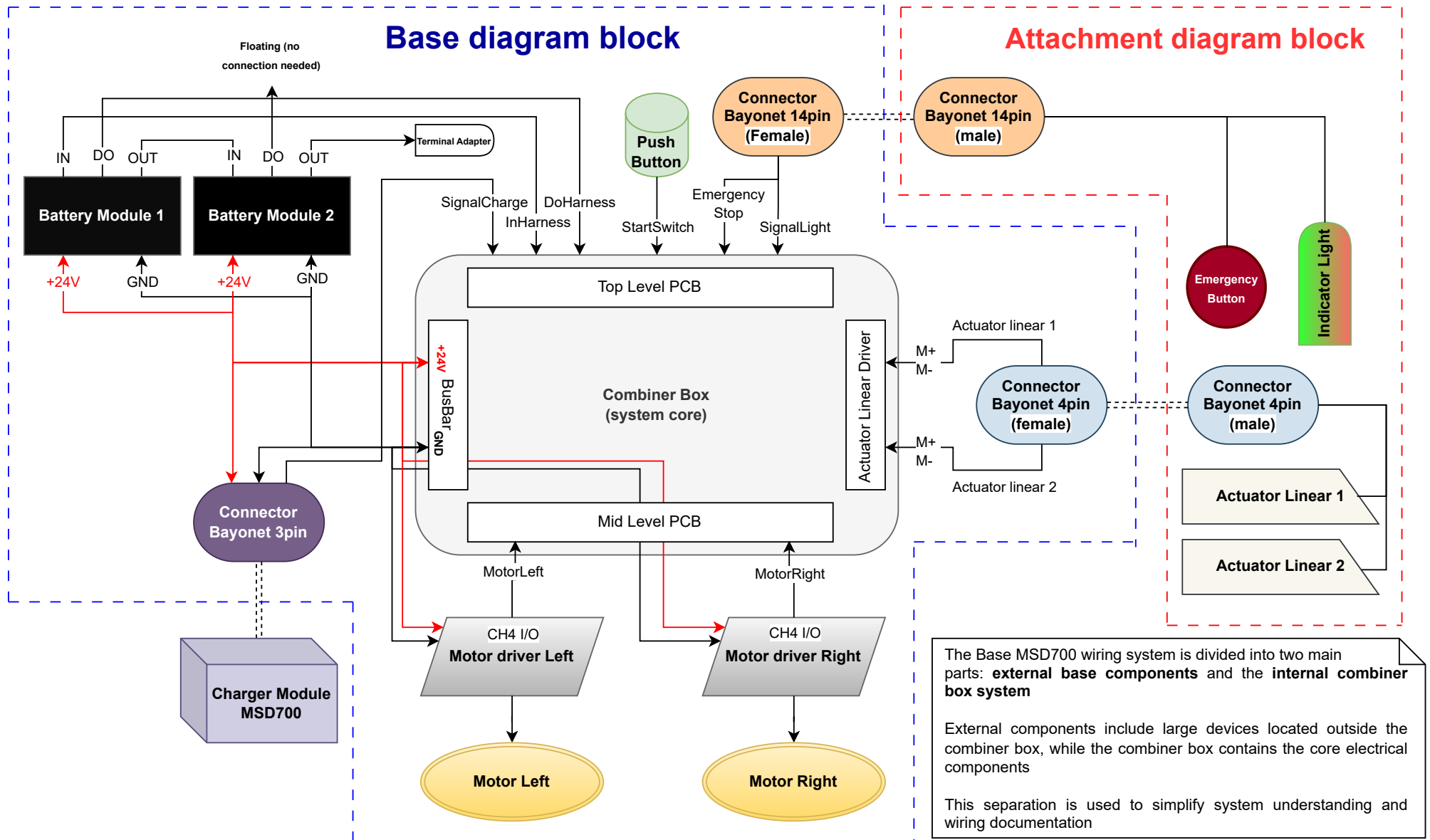
- Components and connection points are marked with numbered labels.
- Clicking a numbered label (**yellow label**) will direct the to a more detailed diagram or explanation on the following pages

General System Overview

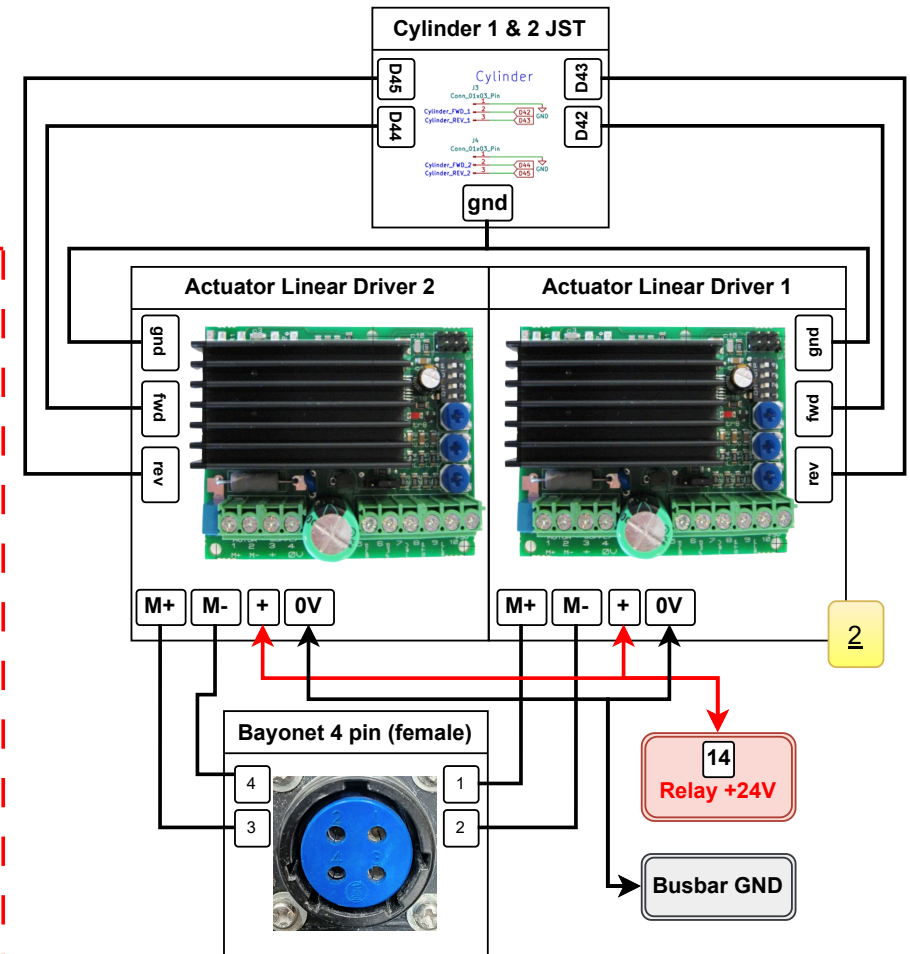
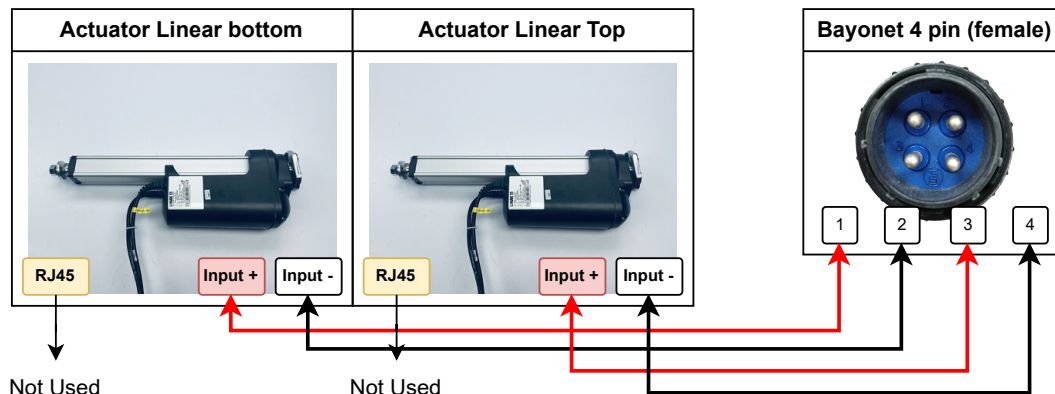
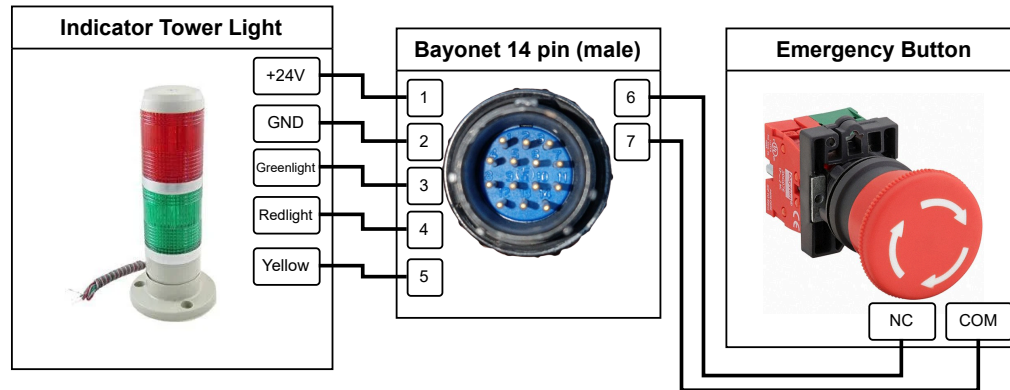
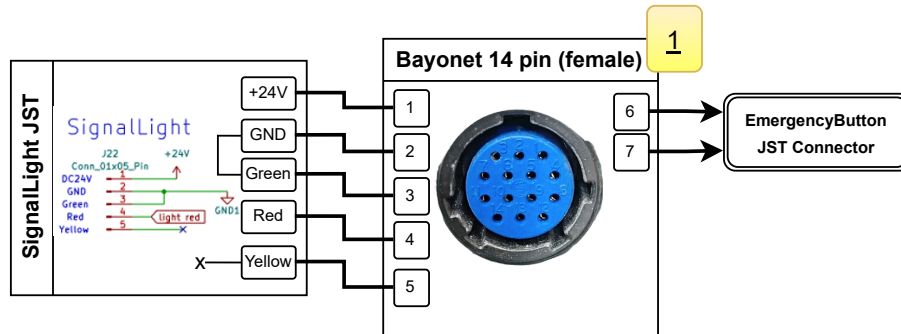
The MSD700 robot system can be divided into two main wiring groups:

1. Base MSD700 Wiring System
2. Attachment Arm Wiring System

The block diagram in this section presents a high-level functional overview using simplified geometric shapes to represent major subsystems without detailing individual pin connections.



Detailed wiring of the Attachment Arm subsystem

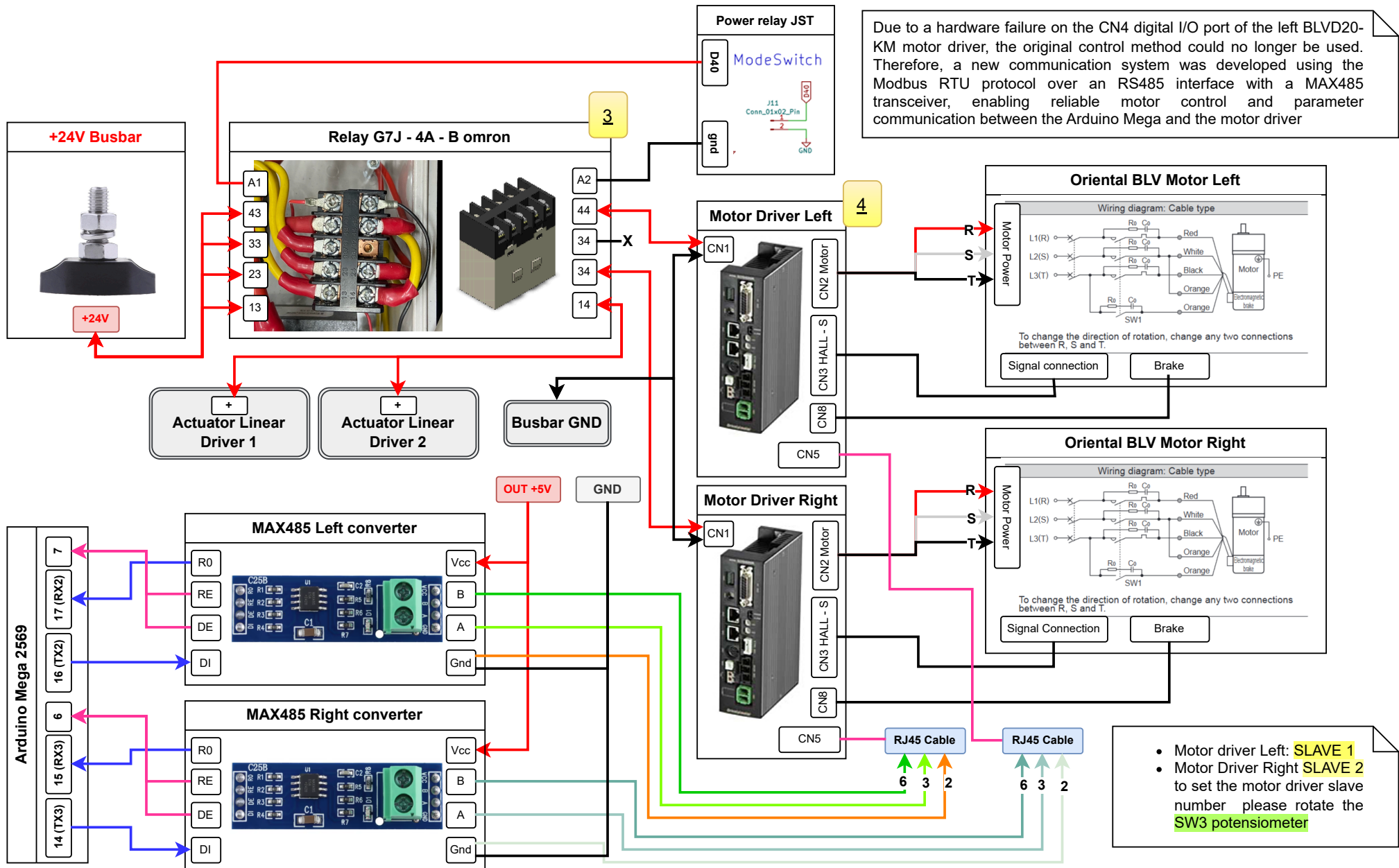


The connection between the Base system and the Attachment Arm is made using a multipin bayonet connector

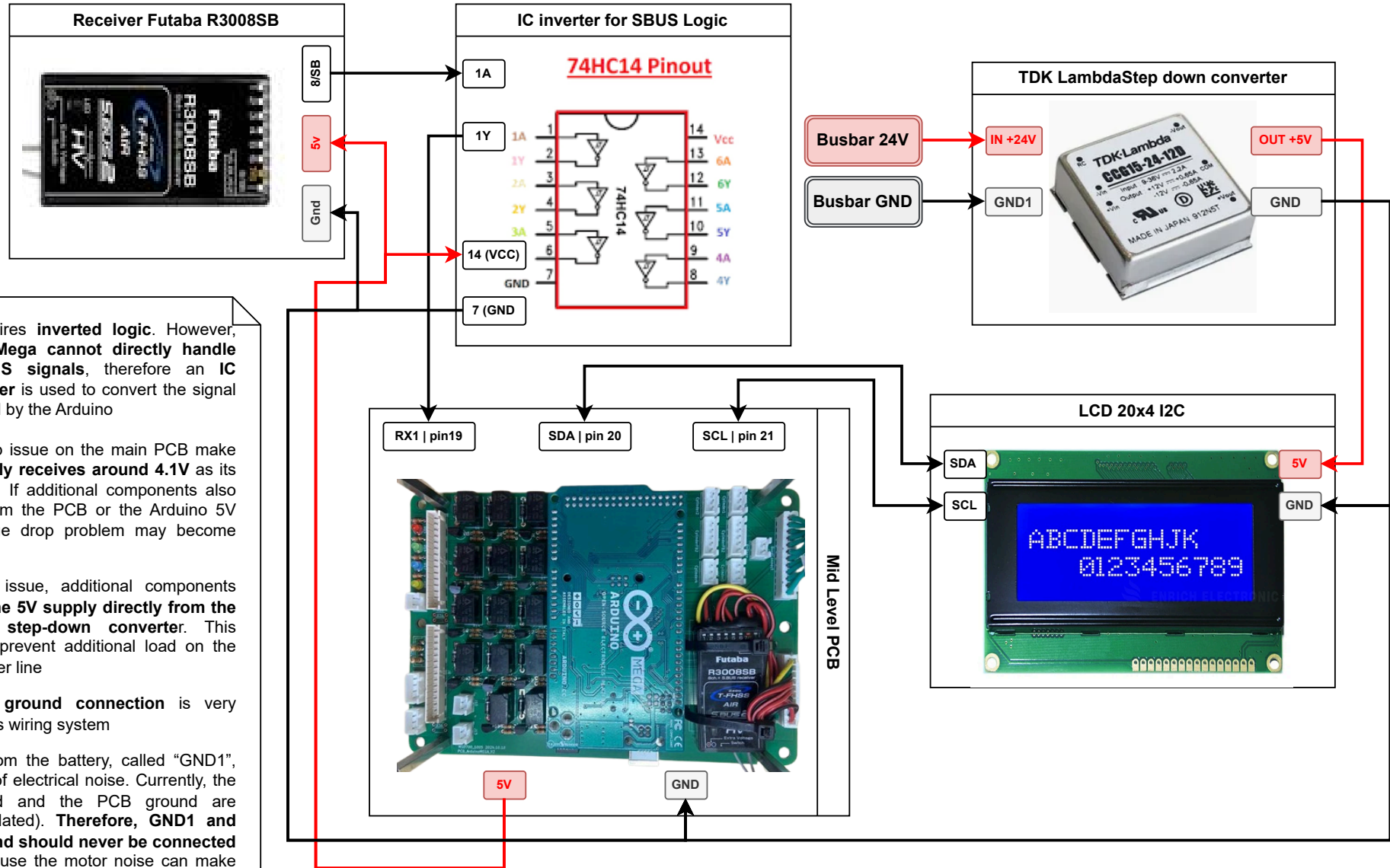
This connector carries multiple power and control signals through different pins, allowing the attachment arm to be easily connected or disconnected from the base system

The pin functions are defined according to the wiring shown in the diagram

Detailed wiring of motor driver BLVD20KM and Omron Relay



Detailed wiring of SBUS interface for receiver RC and LCD



1. SBUS requires **inverted logic**. However, the **Arduino Mega cannot directly handle inverted SBUS signals**, therefore an **IC 74HC14 inverter** is used to convert the signal before it is read by the Arduino

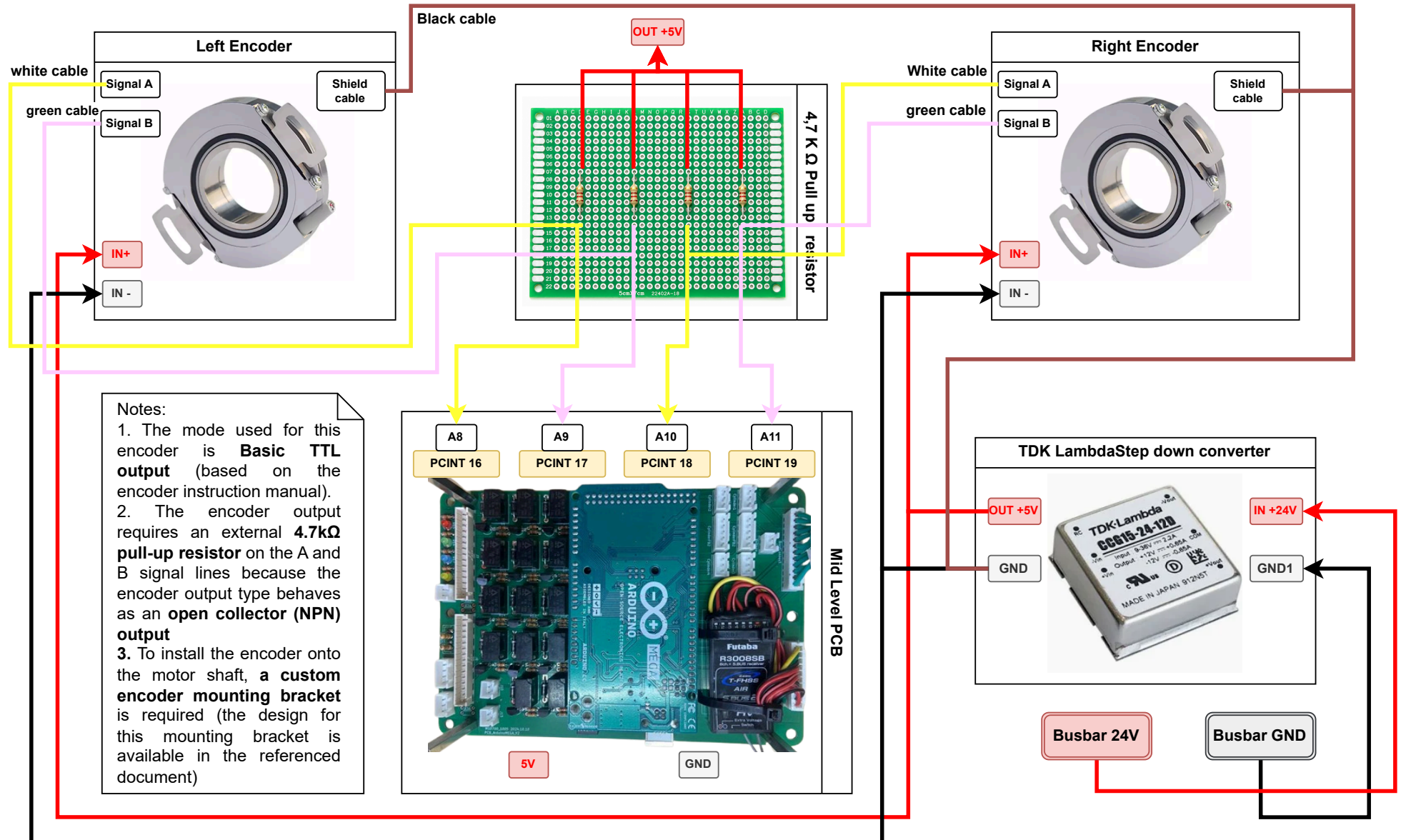
2. Voltage drop issue on the main PCB make the Arduino **only receives around 4.1V** as its supply voltage. If additional components also take power from the PCB or the Arduino 5V pin, the voltage drop problem may become worse

To solve this issue, additional components **should take the 5V supply directly from the TDK-Lambda step-down converter**. This method helps prevent additional load on the main PCB power line

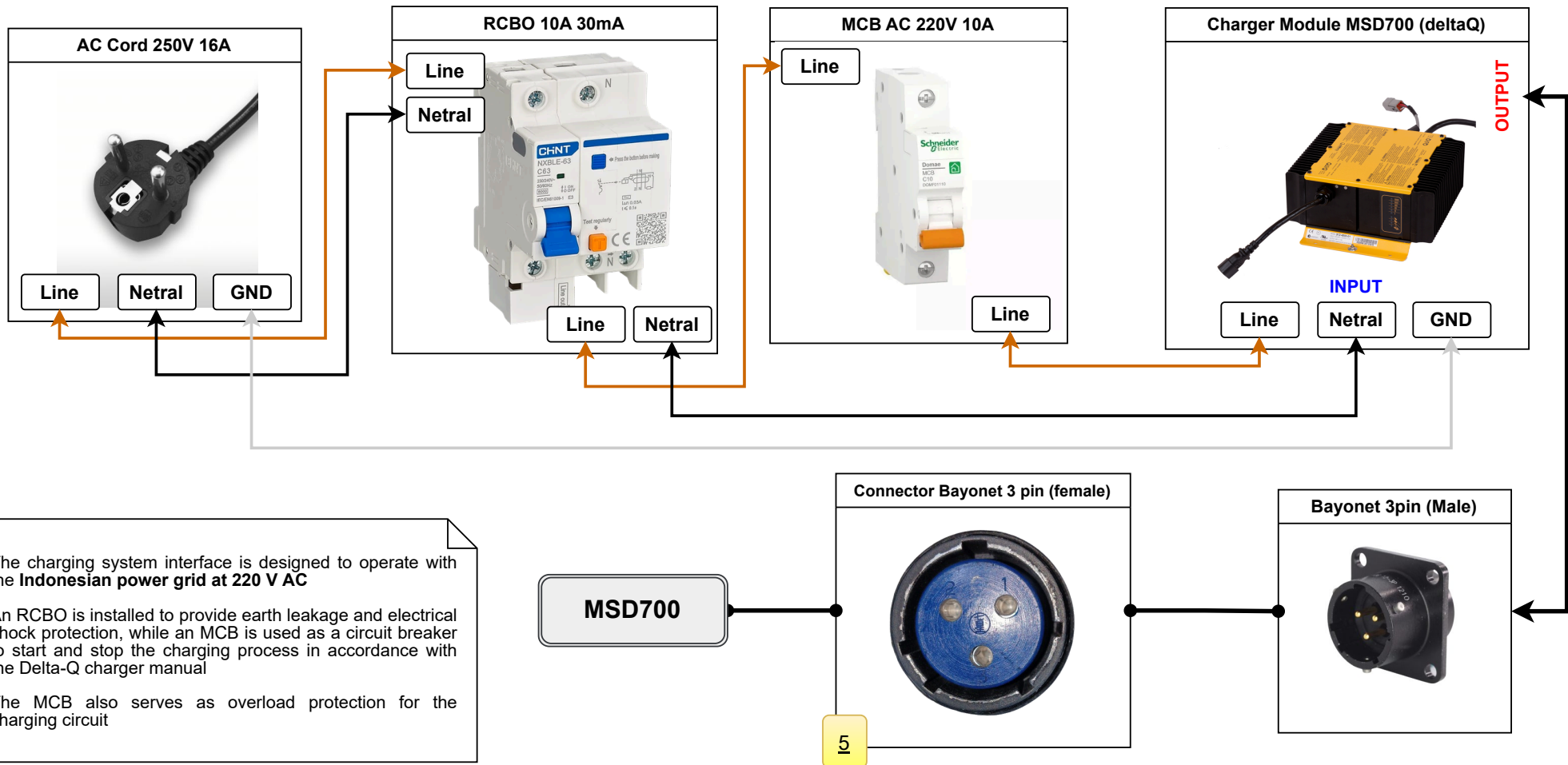
3. Separated **ground connection** is very important in this wiring system

The ground from the battery, called "GND1", contains a lot of electrical noise. Currently, the battery ground and the PCB ground are separated (isolated). **Therefore, GND1 and the PCB ground should never be connected together**, because the motor noise can make the system unstable

Detailed Wiring Diagram of Encoder Interface and Odometry System



Wiring diagram of the charger module interface (220V Indonesia power Grid)



The charging system interface is designed to operate with the **Indonesian power grid at 220 V AC**

An RCBO is installed to provide earth leakage and electrical shock protection, while an MCB is used as a circuit breaker to start and stop the charging process in accordance with the Delta-Q charger manual

The MCB also serves as overload protection for the charging circuit

REFERENCE TO RELEVANT DATASHEETS AND OPERATIONAL NOTES

Component Name	Model/Part Number	Reeference Document
Motor Driver and BLV Motor	BLVD20KM GFS6G100FR	HM-5112_6E
Linear Actuator	LA36	Linerar-actuator-la36
Middle Board and top level Board	(Custom by Nakayama HQ)	ArduinoGithub-FromHQ
Omron Realy and terminal block	Omron G7J-4A-B	G7J-4A-B
Charger Module	DeltaQ (QuiQ 1000)	QuiQ-1000
Battery Module	Toshiba SCiB™ SIP24-23	TOSHIBA Industrial Lithium-ion Battery

REFERENCE TO RELEVANT DATASHEETS AND OPERATIONAL NOTES (2)

Component Name	Model/Part Number	Reference Document
Fuse	Semiconductor Fuse A15QS1004	MERSEN Semiconductor Fuse
Fuse Holder	Ferraz Shawmut P243D	Fuse Block
Stud Terminal Block	TR - S Series	Stud terminal Block TR-S Series
Cable Connector motor driver	CC15D1, General Purpose Cable (1m (3.3ft))	CC15D1
Bayonet Connector	DMS3102CA-18-1-P	Bayonet MS Connector
Receiver Remote Control	Futaba R3008SB	Receiver Futaba R3008

Note: for Bayonet connector, the type used is the same, only the number of pins varies (3 pin, 4 pin, and 14 pin)

REFERENCE TO RELEVANT DATASHEETS AND OPERATIONAL NOTES (3)

Component Name	Model/Part Number	Reference Document
Push Button (Start Switch)	HW1L-A1P11Q4A	22mm Latch Pushbutton Lighted
Emergency Button	XN5E-BV404MR	XN5E-BV404MR
Indicator Tower LED (2 colors)	LR-502-W18	Signal Tower LR 2
Actuator Driver	TR-EM-208	TR-EM_208
Industrial Step Down	Lambda CCG30-24-05S	CCG30-24-05S
Remote Control	Futaba T10J	Futaba-T10J

REFERENCE TO RELEVANT DATASHEETS AND OPERATIONAL NOTES (4)

Component Name	Model/Part Number	Reference Document
Hollow Shaft Encoder	K66-J6C1024Q25	Encoder-K66-J6C1024Q25